

ASE 389P.4 Methods of Orbit Determination

Final project Phase 1

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Throughout, I demonstrate my results with an EKF. In development, I also developed a Batch OD processor and Conventional KF processor, though they struggled to converge.

Assumptions

States

For this phase deliverable, I am only estimating for the following parameters (no biases included yet):

$$\underline{X} = \begin{bmatrix} \underline{r} \\ \underline{\dot{r}} \\ C_D \end{bmatrix}$$

Additionally I've incorporated the following assumptions:

1. Only SRP due to the solar panel.
2. My drag model is full fidelity, incorporating atmospheric position and projected area.
3. Gravity due to 2-Body, Luni-Solar Perturbation, and J2 + J3 + J4 Zonals (NO 20x20 gravity field yet!)
4. I assume no station position bias.

Uncertainties and Numerical Integration Settings

Note that in these uncertainties:

- Uncertainty in the station position is not yet included.
- My process noise uncertainty is likely too low, but I ran into issues when this was set above $10e-8$.

Table 1: Initial Uncertainties and Numerical Integration Settings

Parameter	Symbol	Value	Units
State Vector Uncertainties (Initial P_0)			
Position Std. Dev. (per axis)	σ_{pos}	10	km
Velocity Std. Dev. (per axis)	σ_{vel}	0.01	km/s
Drag Coefficient Std. Dev.	σ_{C_d}	0.05	–
Process Noise (Accel)	σ_{accel}	10^{-11}	km/s ²
Measurement Noise (Station R)			
Range Std. Dev. (Atoll/Arecibo)	σ_ρ	0.01	km
Range Rate Std. Dev. (Atoll/Arecibo)	$\sigma_{\dot{\rho}}$	5×10^{-7}	km/s
Range Std. Dev. (Diego Garcia)	σ_ρ	0.005	km
Range Rate Std. Dev. (Diego Garcia)	$\sigma_{\dot{\rho}}$	10^{-6}	km/s
Numerical Integration Settings (ode45)			
Relative Tolerance	$RelTol$	3×10^{-10}	–
Absolute Tolerance	$AbsTol$	1×10^{-12}	–

Next Steps

Next, I will work on

1. Include range biases in state vector

Knowledge: Fit Interval

Prefit Residuals for All Data and All Sensors

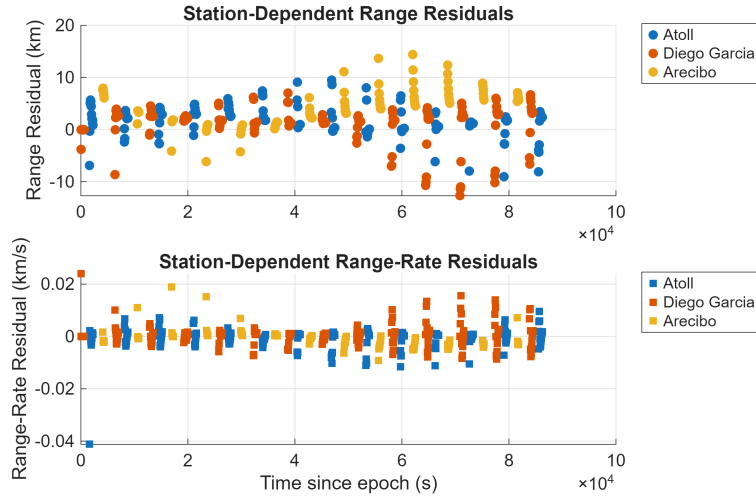


Figure 1: Prefit Residuals using All Data and All Stations

Postfit Residuals for All Data and All Sensors

**Note that because my residuals are still large, I've omitted the 3σ bounds on my plots. These will be added later when they are useful in showing the effectiveness of my filtering.

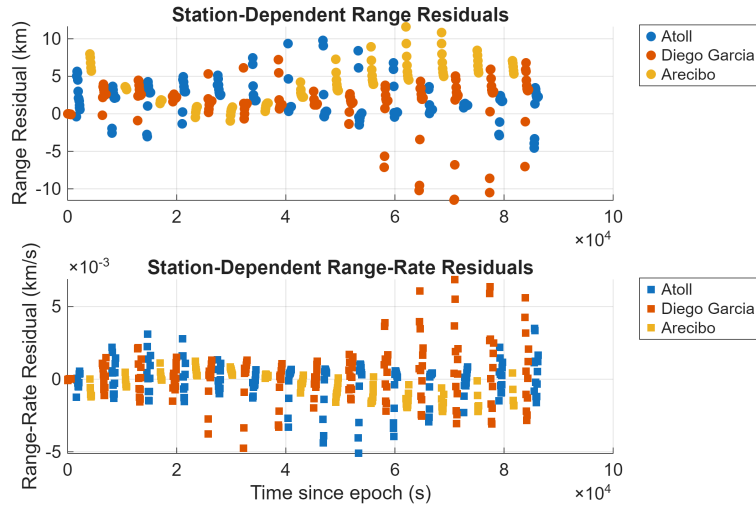


Figure 2: Postfit Residuals using All Data and All Stations

Radial, Intrack, Crosstrack Combination Plots

For this phase, I've includes combinations (a) through (e).

Note that my position discrepancies are so large that my error ellipses are not showing up.

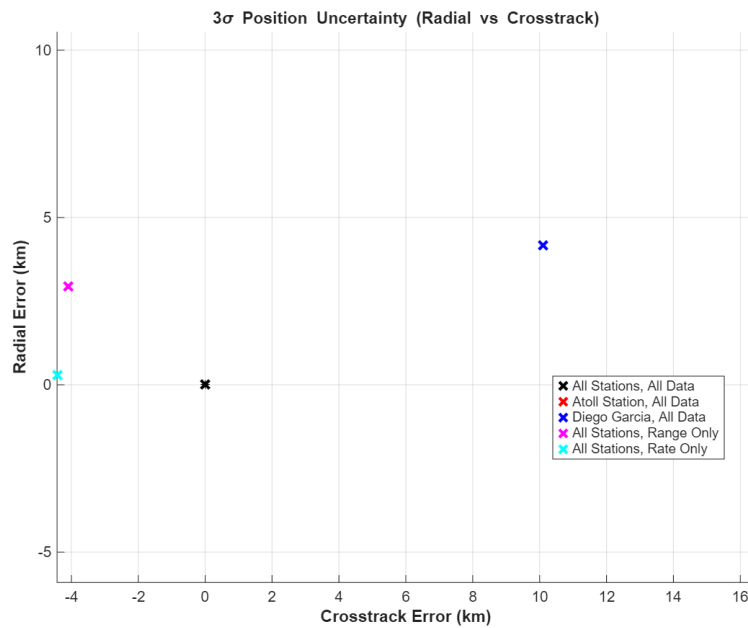


Figure 3: Radial vs Crosstrack

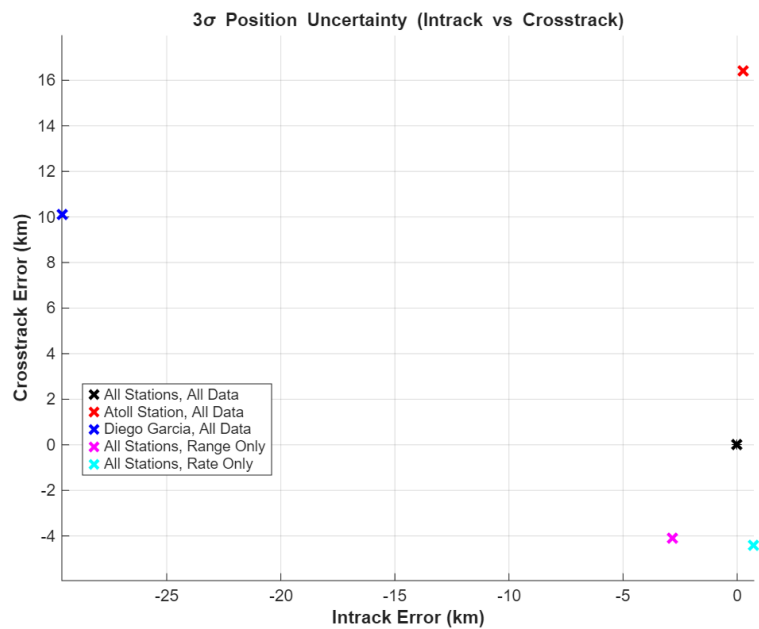


Figure 4: Intrack vs Crosstrack

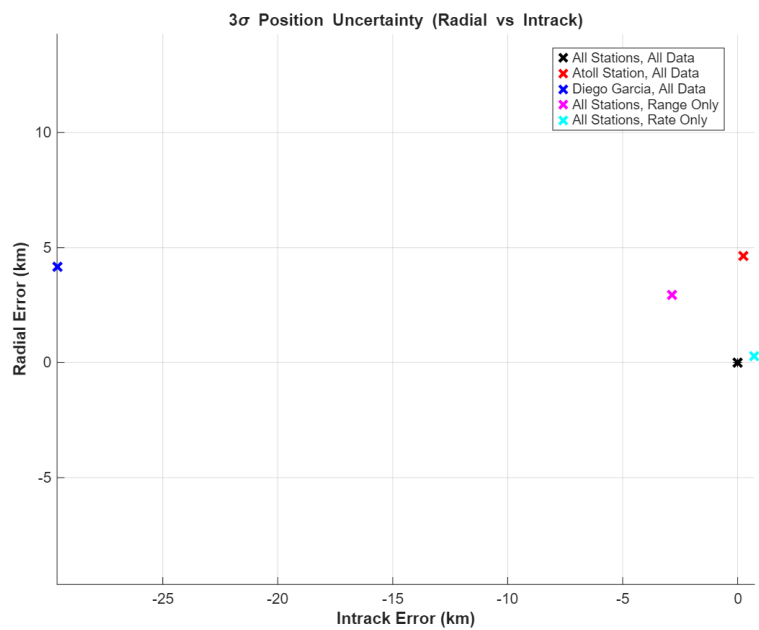


Figure 5: Radial vs Intrack

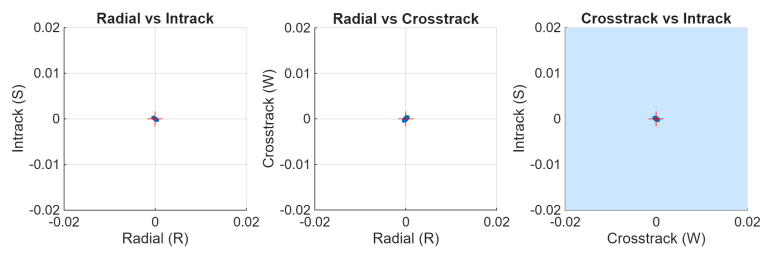


Figure 6: Initial Position RIC Orbit All Sensors All Data

Delivery: Predicted Trajectory

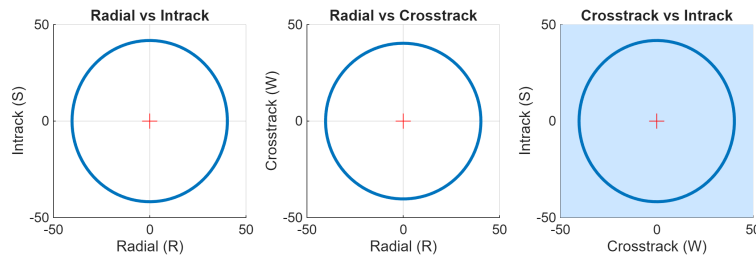


Figure 7: Propagated ΔV_1 for All Sensors and All Data

Discussion

What are the different estimation cases indicating to you physically and statistically?

How does each sensor (station) and data type (range vs. range-rate) contribute to the overall uncertainty, both in knowledge and delivery?

Did you need to compensate for unmodeled dynamics (e.g., process noise, empirical accelerations)? If so, what was your approach and rationale?

Yes, my dynamics model does not yet include the 20x20 gravity field. Thus, I had to compensate with significant process noise. My approach is not very rationale – I arbitrarily chose the magnitude.

How well does your estimated trajectory predict future measurements (i.e., what do the prefit residuals look like for data not used in the fit)?

Not well. My next step is to update and improve my dynamics model to incorporate the 20x20 gravity field.

Did you infer the presence of measurement biases? If so, for which stations and data types, and how did you handle them?

No, not yet.

What states and physical parameters did you *estimate*? What did you *consider*? (A *considered* parameter, in the Schmidt-Kalman sense, is one whose uncertainty is accounted for in the covariance propagation but which is not itself updated by the filter — its effect on the state uncertainty is carried through the cross-covariance without estimating the parameter directly.)

I estimated all of the states in my vector.

$$\underline{X} = \begin{bmatrix} \underline{r} \\ \underline{\dot{r}} \\ C_D \end{bmatrix}$$

However, I should be just marking C_D as a *consider* parameter. In my initial efforts with both a Batch OD and Conventional KF, the drag coefficient blew up to compensate for my poor modeling. The EKF seemed more robust, but was not able to provide any useful information.

How observable are your estimated states and parameters given the tracking geometry?

Not yet investigated.

Did you smooth the trajectory? If so, how did this improve your solution?

Not yet performed.

What was your overall strategy for arriving at a converged, consistent solution?

So far, the strategy is simply to build a basic EKF structure with my base dynamics model. My solution is not yet converged nor consistent.

1 Code Appendix

All code can be found on this GitHub repository: https://github.com/Isaac-BYU-UT/Methods_OD_Final_Project